

Advanced Communication Architectures (TSM_AdvComArc)

Cloud Services & Network Virtualization

Cloud and Network Architecture Evolution

Objectifs

- After the course, participants will be able to:
 - Explain the concepts of « Network Virtualisation »
 - Explain the different services offered by « Cloud » solutions
 - Describe the general SDN architecture and its main functional components
 - Describe functionality and placement of OpenFlow in a SDN solution
 - Explain OpenFlow functions and mechanisms
 - Locate SDN place in a virtualization architecture

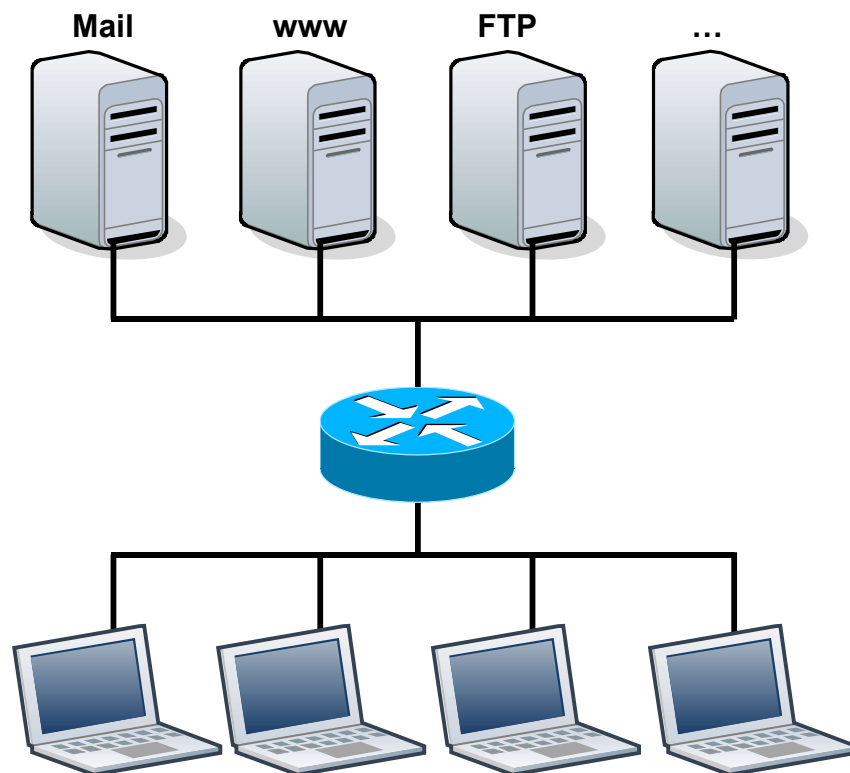
Content

- Introduction:
 - Network architecture evolution summary
 - Data center
- Cloud
 - Cloud services
 - Cloud models
- Network Virtualization
 - Network Virtualization: State of the Art
 - Network Function Virtualization
 - Virtual network components
- Cloud networking
- VM Migration

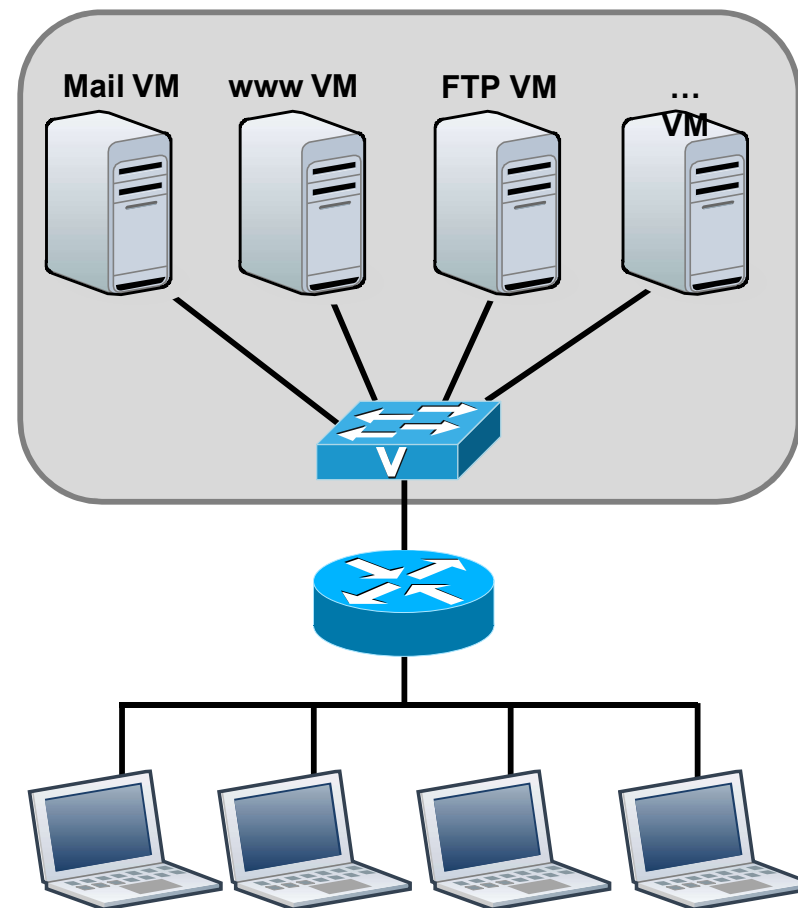


Network architecture evolution

Standalone servers

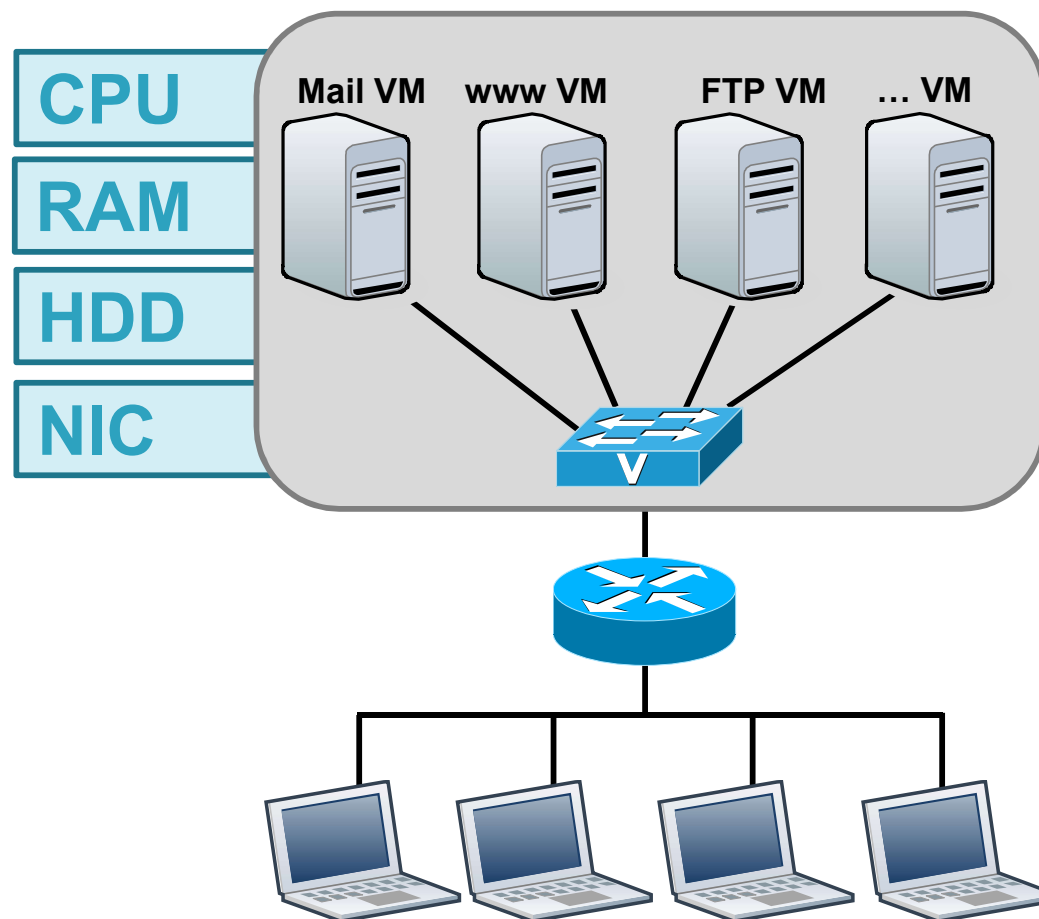


Server virtualization

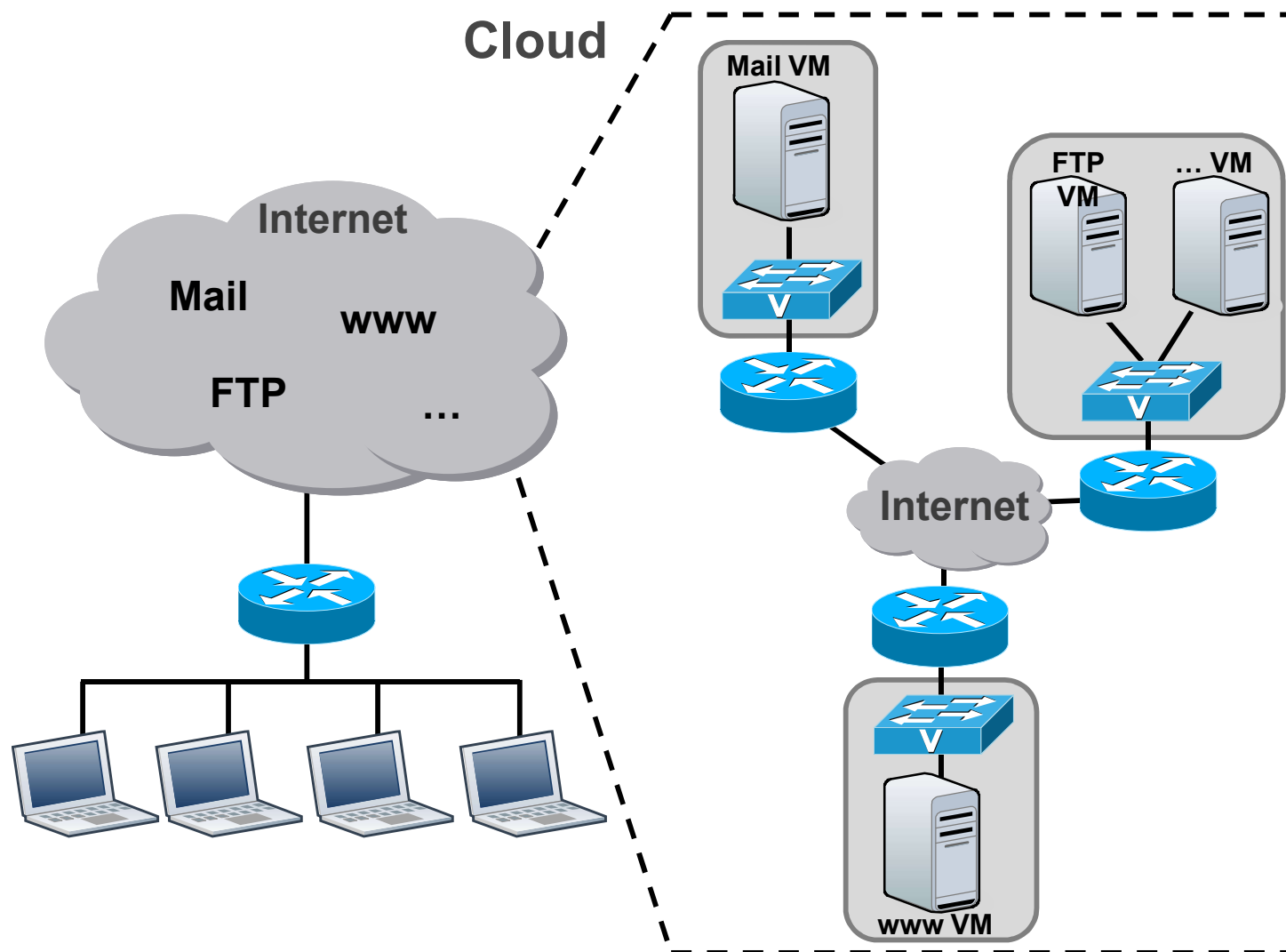


Network architecture evolution

Data center



Network architecture evolution



Data center – Definition

- “A data center is a facility used to house computer systems and associated components, such as **telecommunications, computing and storage systems**.
- It generally includes redundant or backup power supplies, redundant data communications connections, environmental controls (e.g., air conditioning, fire suppression, power continuity infrastructures) and security devices.

Wikipedia: https://en.wikipedia.org/wiki/Data_center

Data center (1)

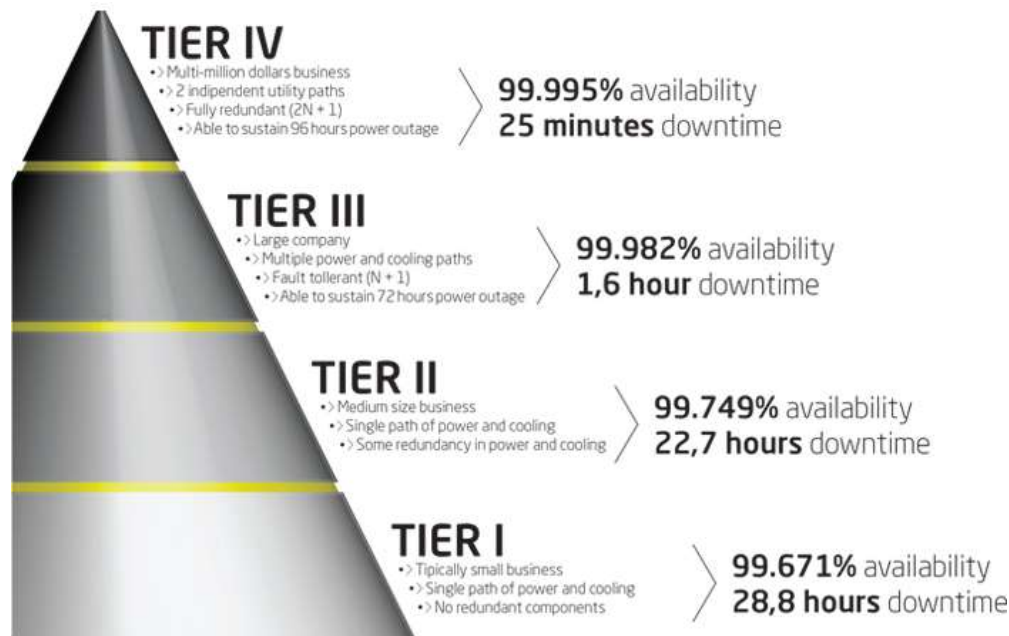


Tier definition

The Tier certifications are awarded in four levels:

- Tier IV - Fault tolerant site infrastructure
- Tier III - Concurrently maintainable site infrastructure
- Tier II - Redundant capacity components site infrastructure (redundant)
- Tier I - Basic site infrastructure (non-redundant)

Interesting link: <https://www.youtube.com/user/uptimeinstitute>



Tier definition

FEATURES	TIER I	TIER II	TIER III	TIER IV
Single non-redundant distribution path serving the IT equipment	✓	✓	✓	✓
Non-redundant capacity components	✓	✓	✓	✓
Basic site infrastructure with expected availability of 99.671%	✓	✓	✓	✓
Redundant site infrastructure capacity components with expected availability of 99.741%	-	✓	✓	✓
Multiple independent distribution paths serving the IT equipment	-	-	✓	✓
All IT equipment must be dual-powered and fully compatible with the topology of a site's architecture	-	-	✓	✓
Concurrently maintainable site infrastructure with expected availability of 99.982%	-	-	✓	✓
All cooling equipment is independently dual-powered, including chillers and heating, ventilating and air-conditioning (HVAC) systems	-	-	-	✓
Fault-tolerant site infrastructure with electrical power storage and distribution facilities with expected availability of 99.995%	-	-	-	✓

<https://www.chintglobal.com/global/en/about-us/news-center/blog/how-to-choose-data-center.html>

Data center (2)

A couple of interesting links:

- YouTube: Explore a Google data center
<https://www.youtube.com/watch?v=avP5d16wEp0>
- Street View: Explore a Google data center
<https://datacenters.google/>
- Security layers at Google Datacenters:
<https://www.youtube.com/watch?v=kd33UVZhnAA&t=84s>



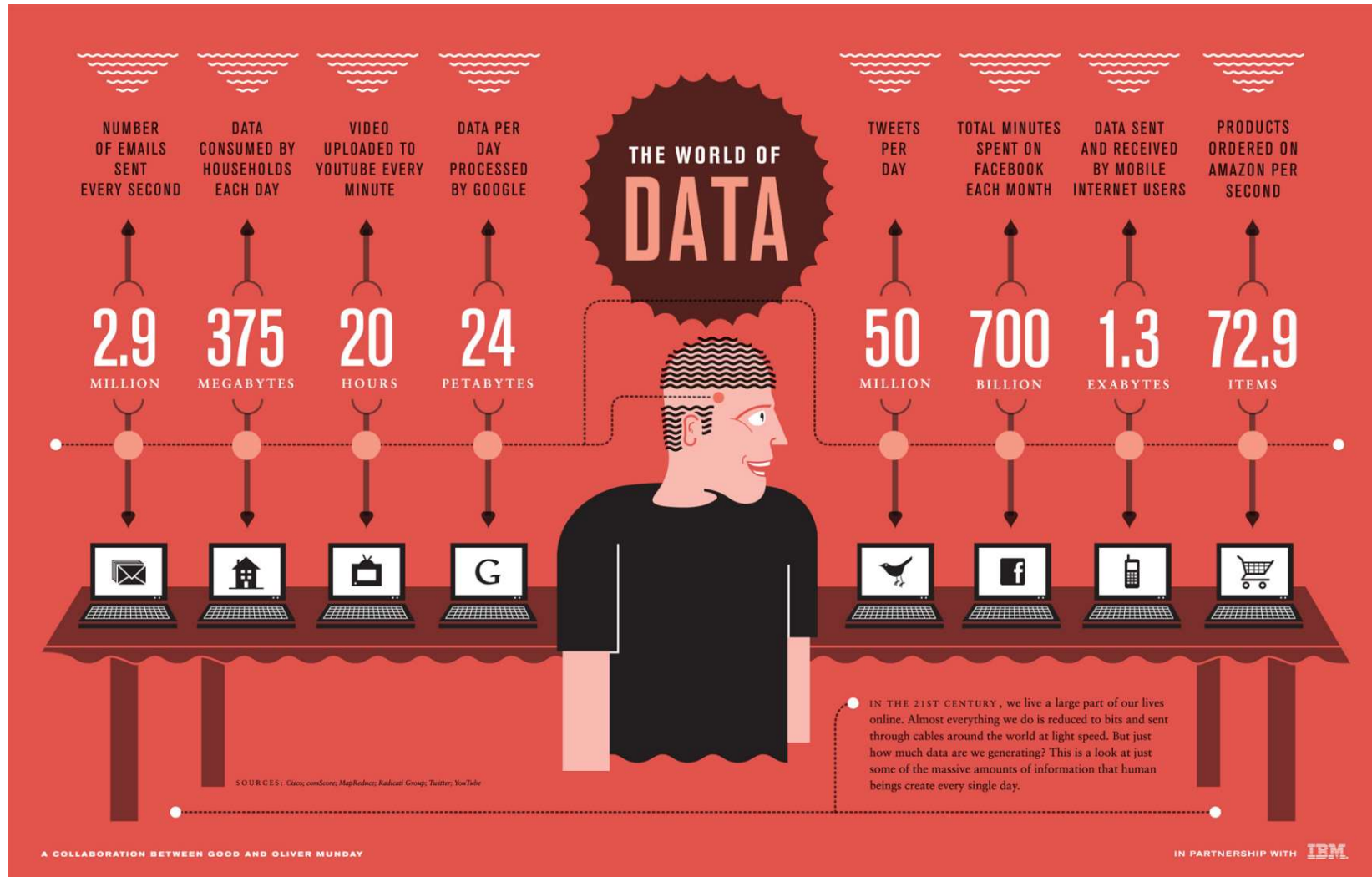
Introduction – Glossary

Glossary

- <https://www.techtarget.com/whatis/glossary/Data-center-management>
- <https://whatis.techtarget.com/glossary/Storage-Hardware>
- <https://www.techtarget.com/searchnetworking/definition/network-fabric>
- <https://www.ninjaone.com/blog/common-network-issues/>

Cloud

The world of DATA



Data volumes around the Zettabyte (10^{21} bytes)

Units of Computer Memory Measurements

1 Bit	= Binary Digit
8 Bits	= 1 Byte
1024 Bytes	= 1 KB [Kilo Byte]
1024 KB	= 1 MB [Mega Byte]
1024 MB	= 1 GB [Giga Byte]
1024 GB	= 1 TB [Terra Byte]
1024 TB	= 1 PB [Peta Byte]
1024 PB	= 1 EB [Exa Byte]
1024 EB	= 1 ZB [Zetta Byte]
1024 ZB	= 1 YB [Yotta Byte]
1024 YB	= 1 Bronto Byte
1024 Brontobyte	= 1 Geop Byte

Evolution of water Utility



Source: M.F. Sakr, CMU Qatar

Transformation of IT from a Product to a Service



Source: M.F. Sakr, CMU Qatar

Cloud computing – Definition

- “Cloud computing refers to **applications** and **services** offered over the Internet”.
- These services are offered from data centers all over the world, which collectively are referred to as the “cloud.”
- This metaphor represents the intangible, yet universal nature of the Internet.

Techterms: http://www.techterms.com/definition/cloud_computing

Why Cloud computing ?

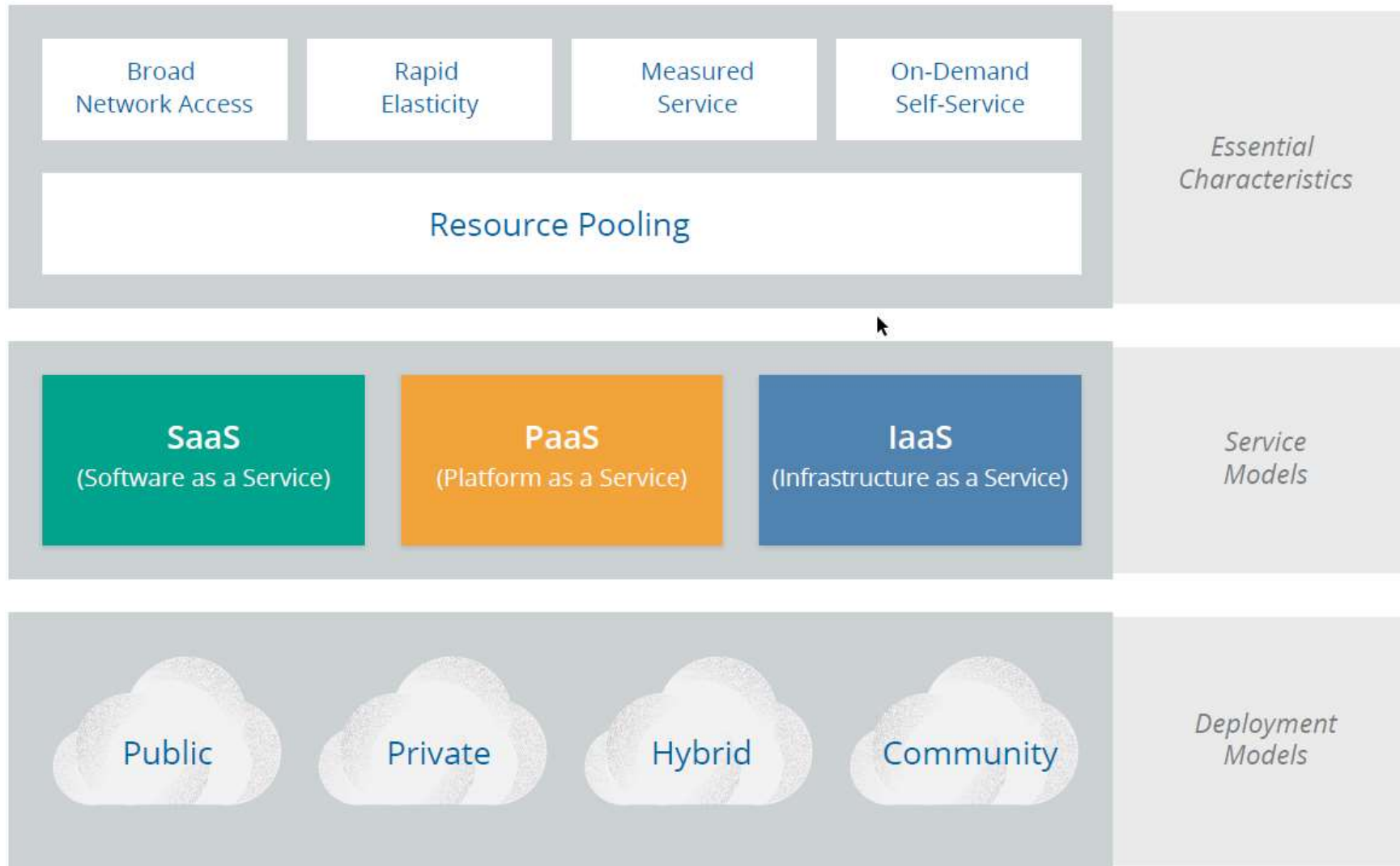


Source: M.F. Sakr, CMU Qatar

Characteristics of Cloud Solutions

- Resource pooling is the most fundamental characteristic, as discussed above. The provider abstracts resources and collects them into a pool, portions of which can be allocated to different consumers (typically based on policies).
- Consumers provision the resources from the pool using on-demand self-service, managing resources themselves, without having to talk to a human administrator.
- Broad network access means that all resources are available over a network, without any need for direct physical access; the network is not necessarily part of the service.
- Rapid elasticity allows consumers to expand or contract the resources they use from the pool (provisioning and deprovisioning), often completely automatically. This allows them to more closely match resource consumption with demand (for example, adding virtual servers as demand increases, then shutting them down when demand drops).
- Measured service meters what is provided, to ensure that consumers only use what they are allotted, and, if necessary, to charge them for it. This is where the term utility computing comes from, since computing resources can now be consumed like water and electricity, with the client only paying for what they use.

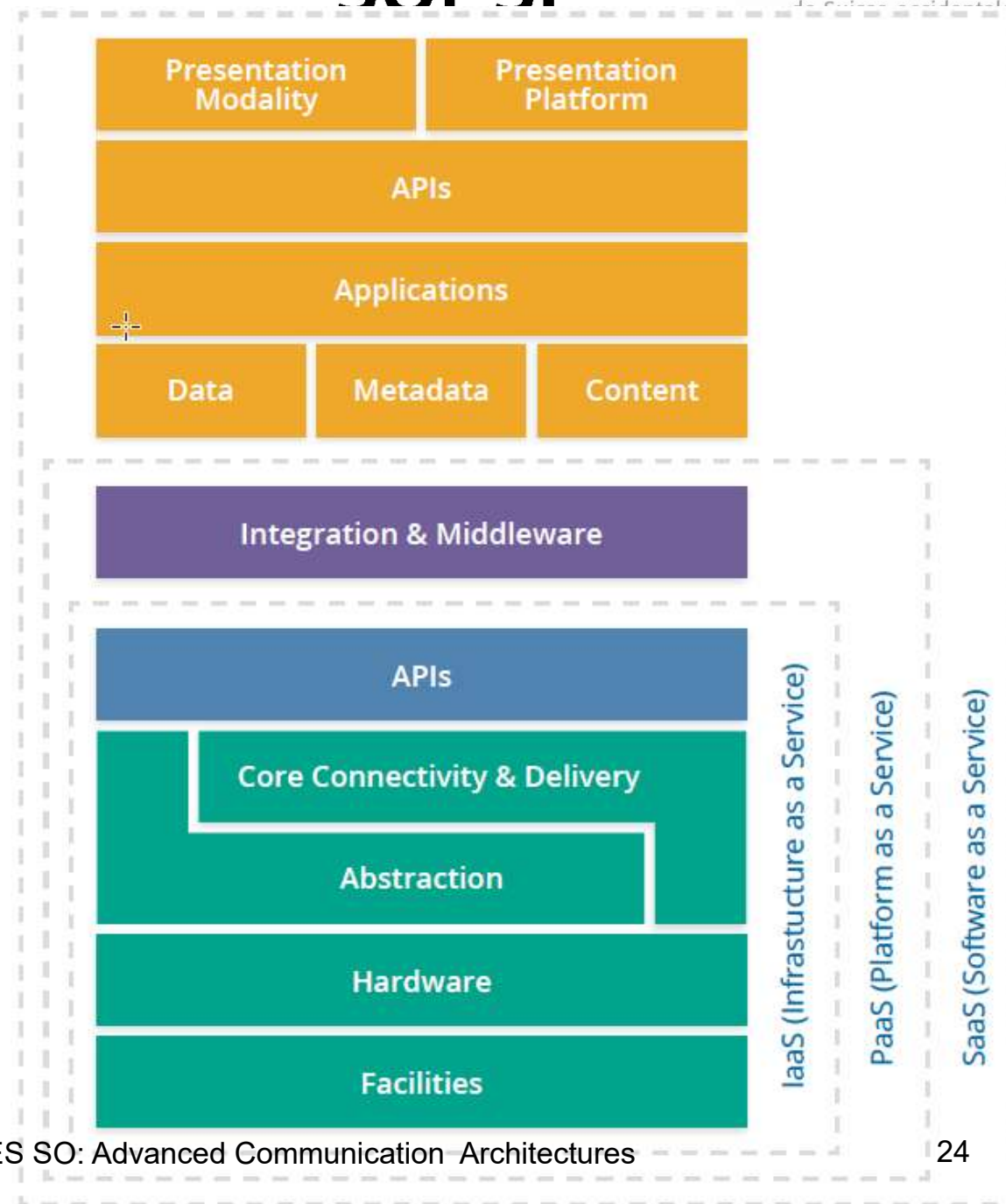
Cloud services



Cloud services

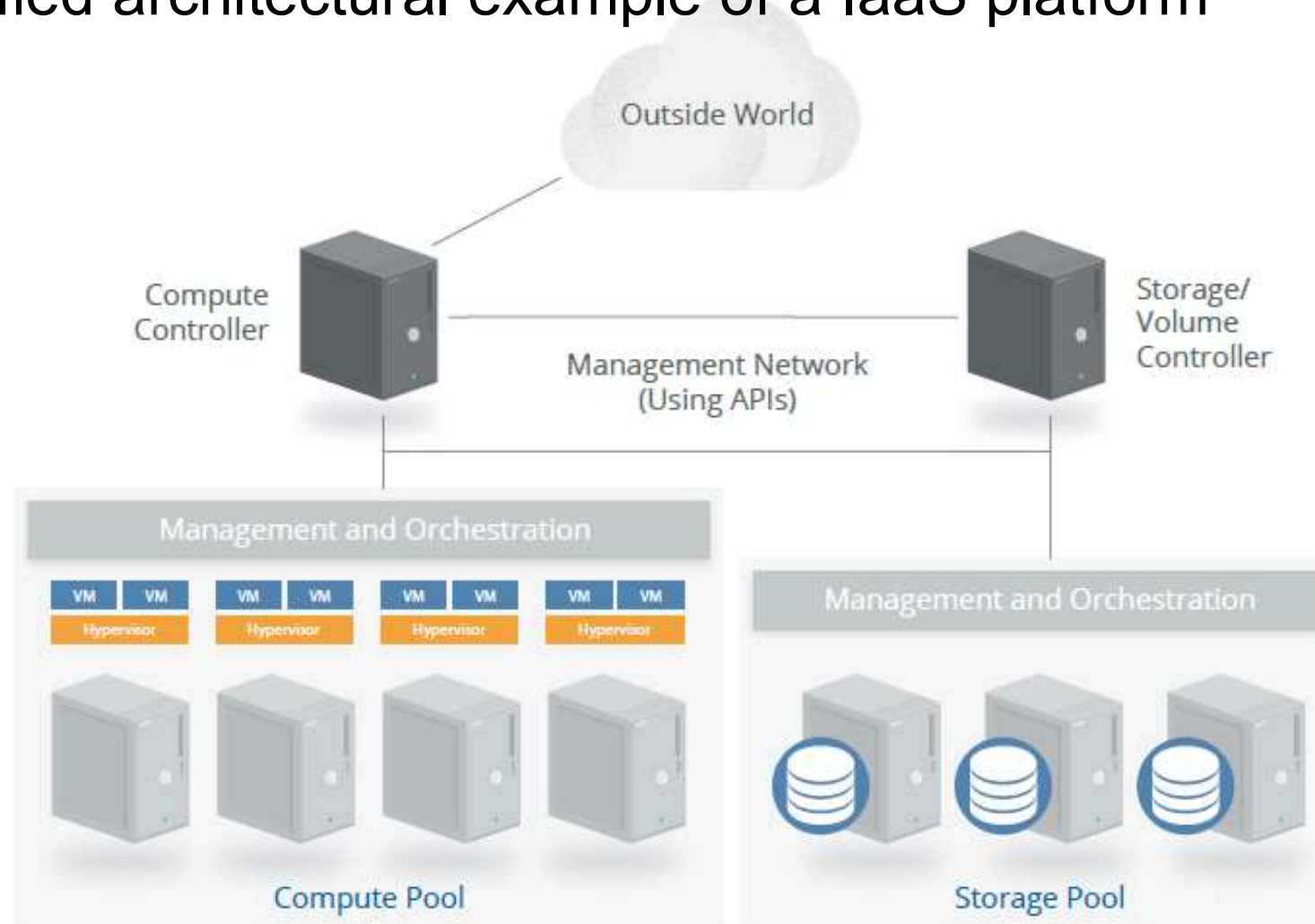
- Software as a Service (SaaS) is a full application that's managed and hosted by the provider. Consumers access it with a web browser, mobile app, or a lightweight client app.
- Platform as a Service (PaaS) abstracts and provides development or application platforms, such as databases, application platforms (e.g. a place to run Python, PHP, or other code), file storage and collaboration, or even proprietary application processing (such as machine learning, big data processing, or direct Application Programming Interfaces (API) access to features of a full SaaS application). The key differentiator is that, with PaaS, you don't manage the underlying servers, networks, or other infrastructure.
- Infrastructure as a Service (IaaS) offers access to a resource pool of fundamental computing infrastructure, such as compute, network, or storage.

Cloud services: reference architecture



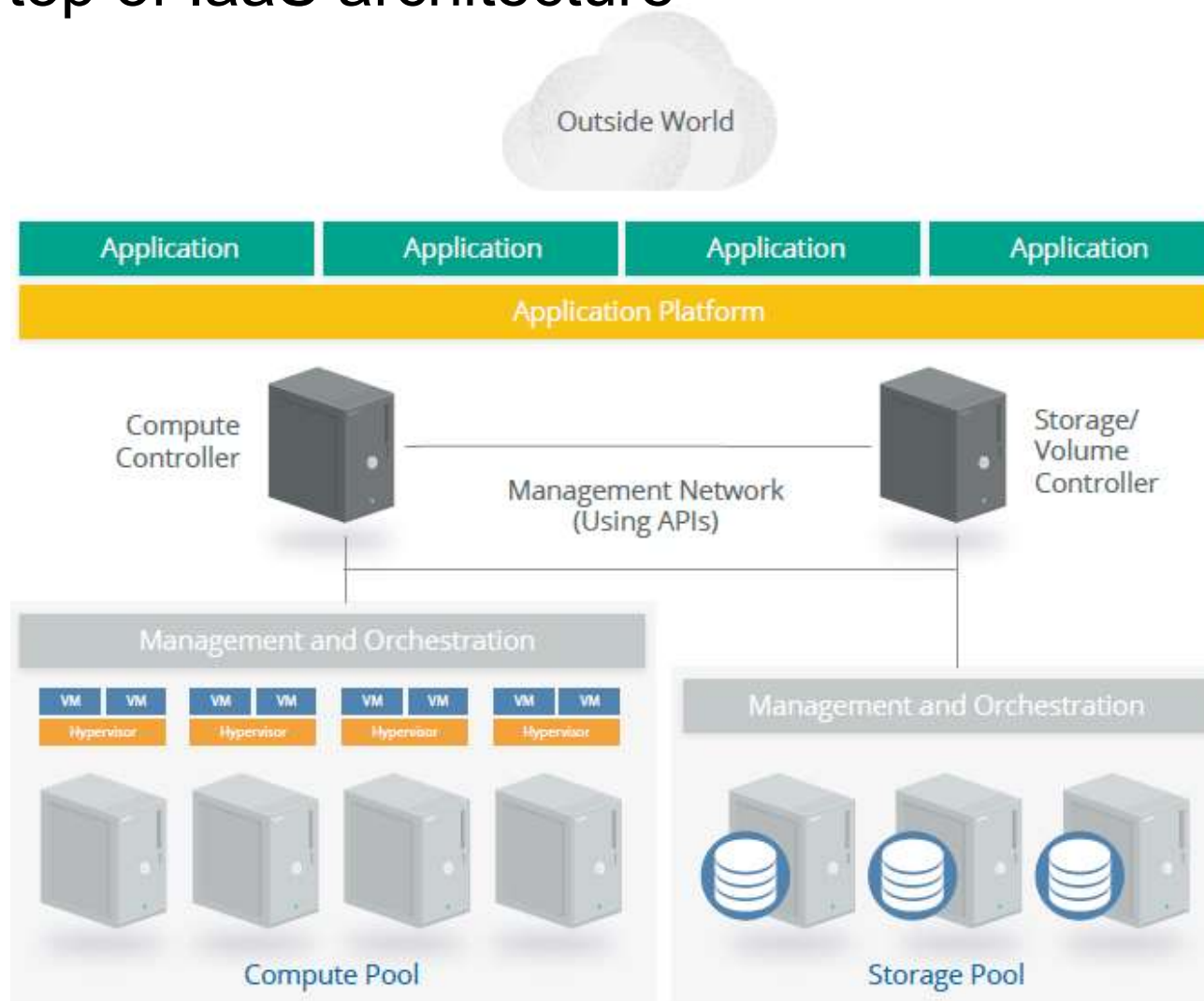
Cloud services

Simplified architectural example of a IaaS platform



Cloud services

Simplified architecture diagram: an application platform (SaaS) running on top of IaaS architecture



Cloud services – IaaS

- **Infrastructure-as-a-Service** is a provision model in which an organization outsources the equipment used to support operations, including storage, hardware, servers and networking components.
- Characteristics and components of IaaS include:
 - Utility computing service and billing model
 - Automation of administrative tasks
 - Dynamic scaling
 - Desktop virtualization
 - Policy-based services
 - Internet connectivity

Cloud services – PaaS

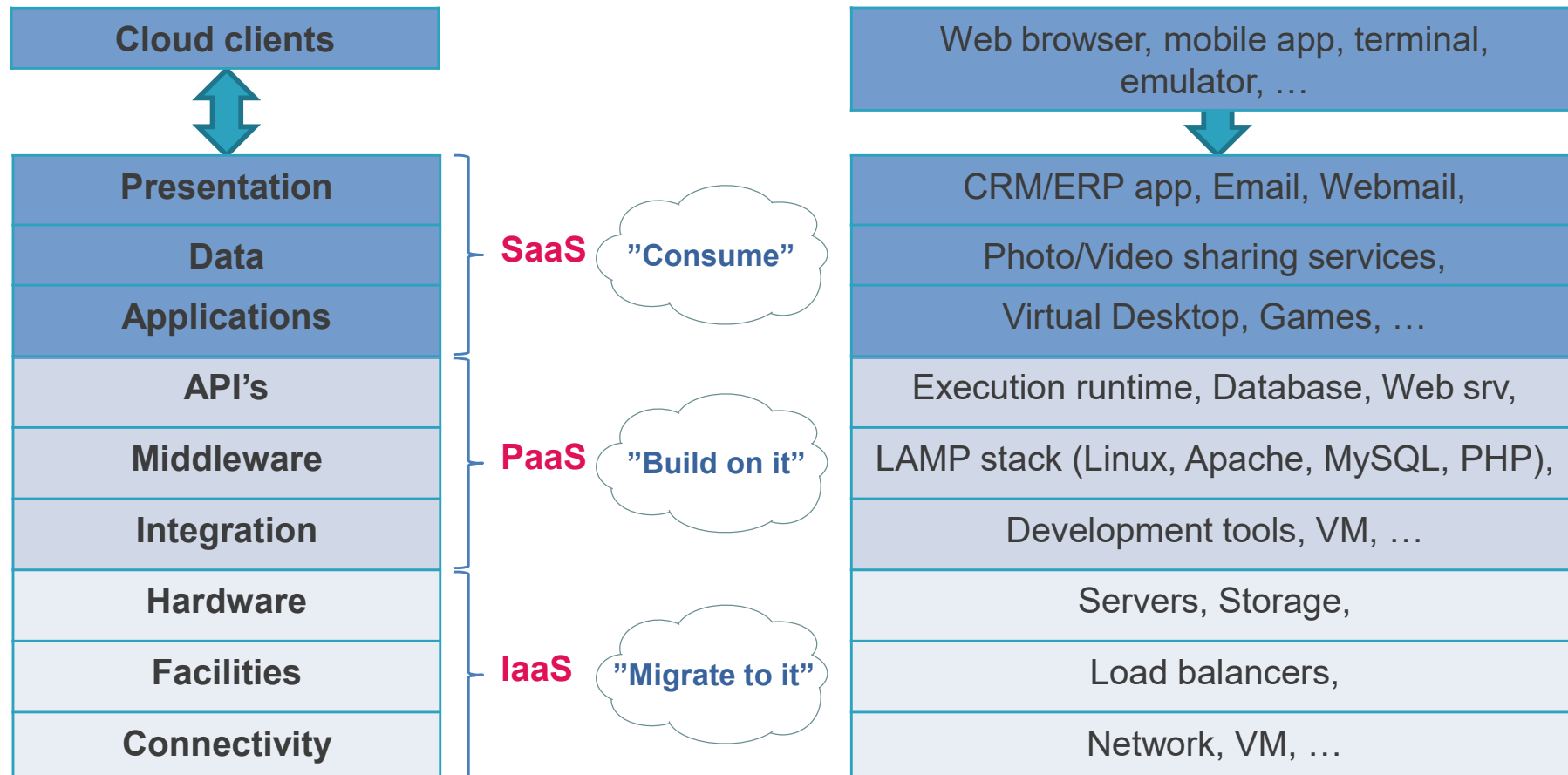
- **Platform-as-a-Service** in the cloud is defined as a set of software and product development tools hosted on the provider's infrastructure.
- Developers create applications on the provider's platform over the Internet. PaaS providers may use APIs, website portals or gateway software installed on the customer's computer. GoogleEngine is an example of PaaS.

Cloud services – SaaS

- **Software-as-a-Service** is a software distribution model in which applications are hosted by a vendor or service provider and made available to customers over a network, typically the Internet.
- Benefits of the SaaS model include:
 - easier administration
 - automatic updates and patch management
 - compatibility: All users will have the same version of software.
 - easier collaboration, for the same reason
 - global accessibility.

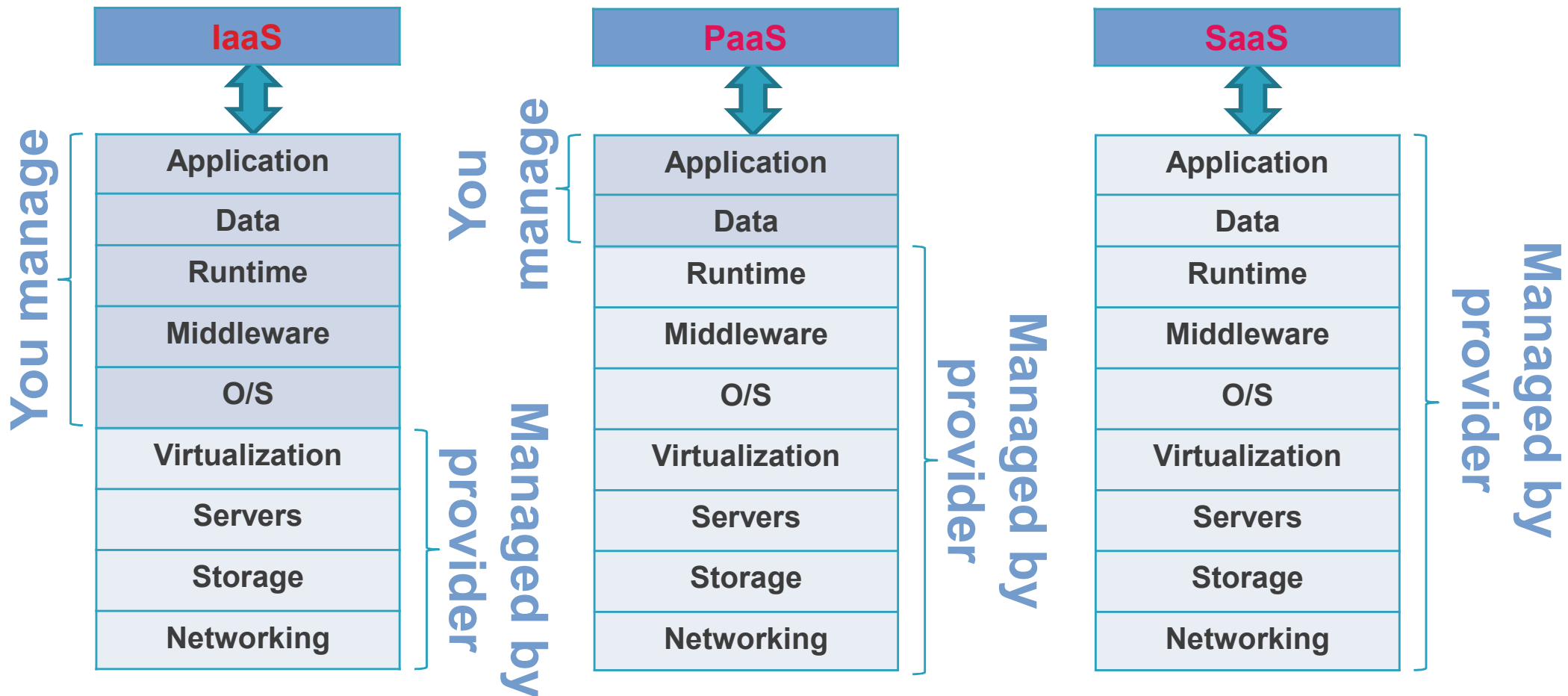
Cloud services – 1

Layer representation of cloud services



Cloud services – 2

Separation of responsibilities



Cloud services – 3

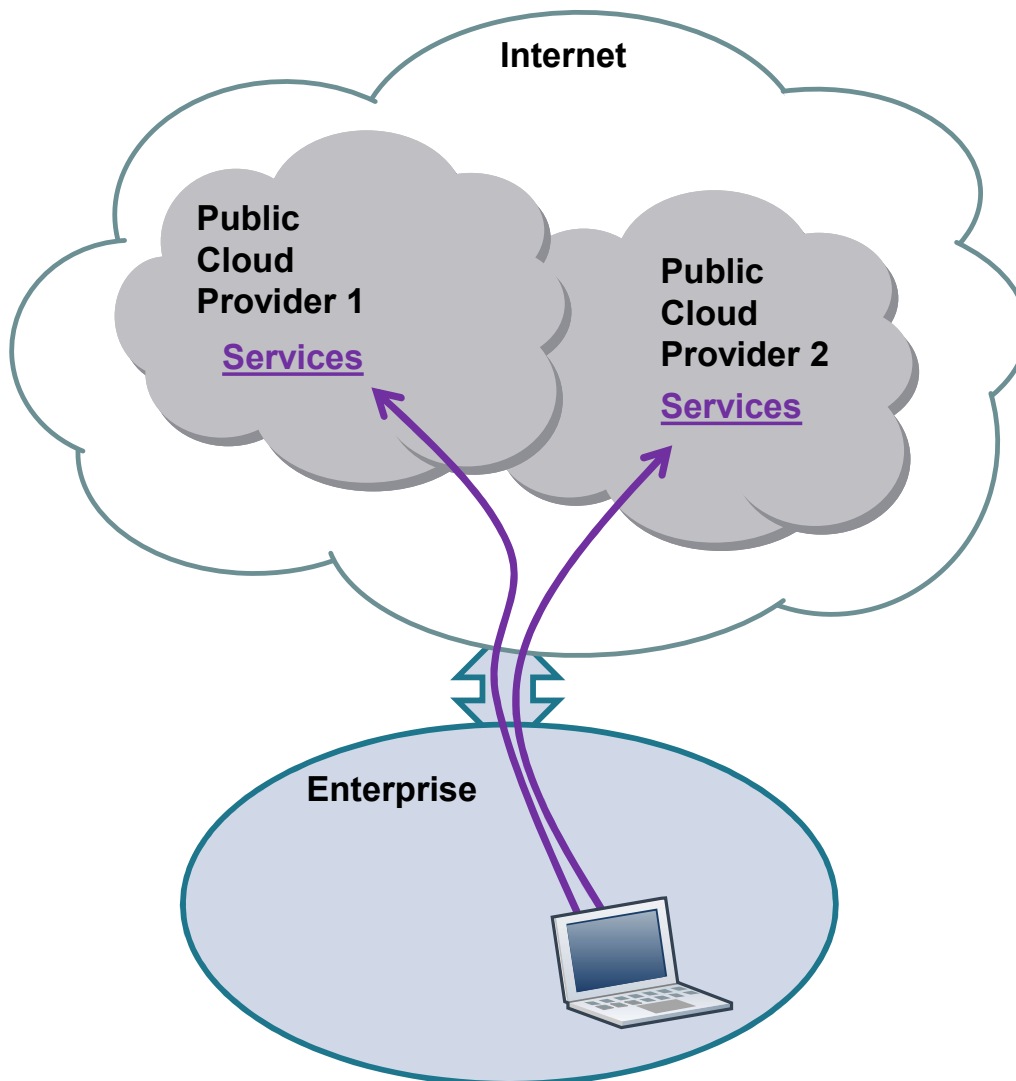
Three distinct characteristics that differentiate a cloud service (IaaS, PaaS, SaaS, ...) from traditional hosting:

- It is **sold on demand** (Typically by the minute or the hour)
- It is **elastic**: a user can have as much or as little of a service as they want at any given time
- The service is **fully managed by the provider**: the consumer needs nothing but a personal computer and Internet access

Cloud services security

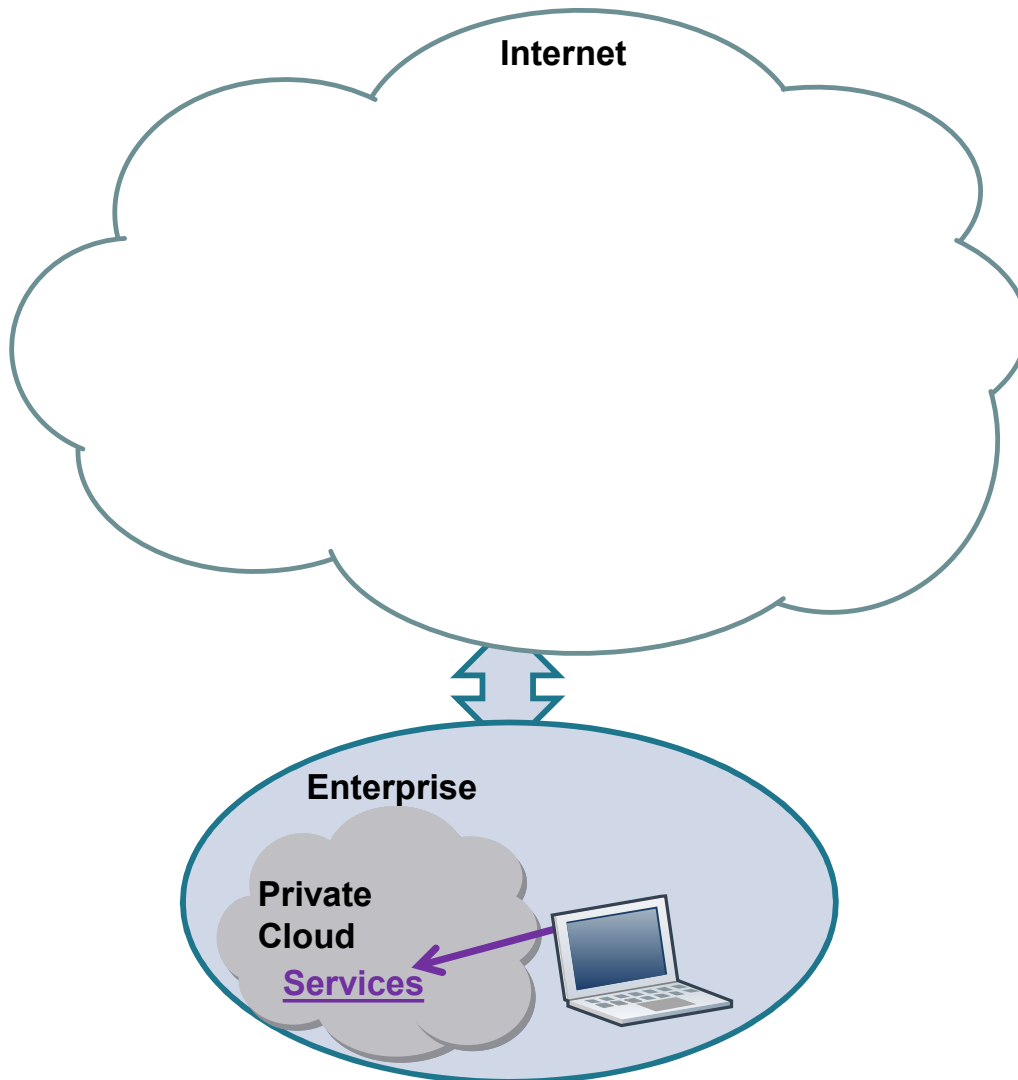
	Infrastructure Owned By ¹	Infrastructure Owned By ²	Infrastructure Located ³	Accessible and Consumed By ⁴
Public	Third-Party Provider	Third-Party Provider	Off-Premises	Untrusted
Private/ Community	Organization Third-Party Provider	Organization Third-Party Provider	On-Premises Off-Premises	Trusted
Hybrid	<u>Both</u> Organization & Third-Party Provider	<u>Both</u> Organization & Third-Party Provider	<u>Both</u> On-Premises & Off-Premises	Trusted & Untrusted

Cloud model – Public cloud



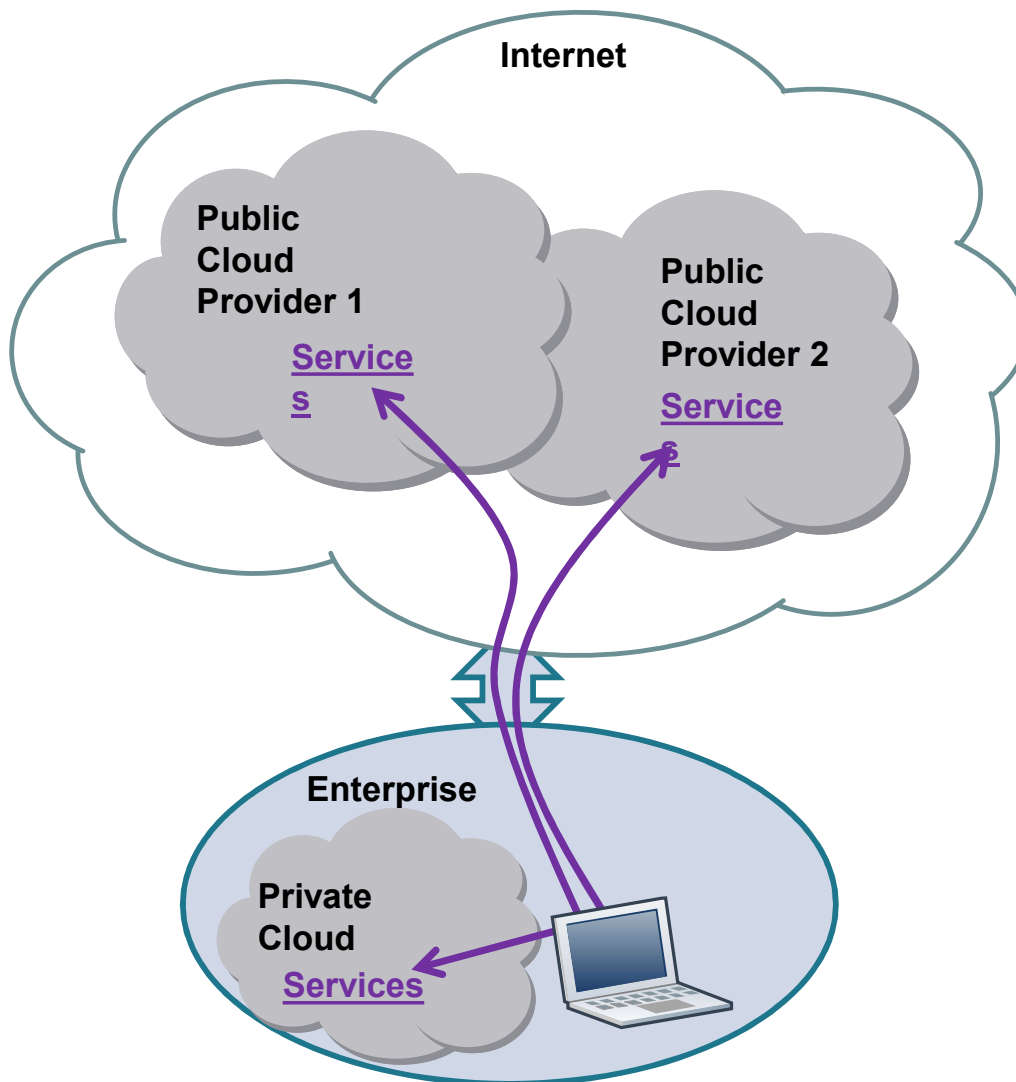
- Resources are available (free or offered on a pay-per-usage model) to the general public over the Internet.
- Examples:
Amazon Elastic Compute Cloud (EC2), IBM's Blue Cloud, Sun Cloud, Google AppEngine and Windows Azure Services Platform.

Cloud model – Private cloud



- Private cloud is a proprietary computing architecture that provides hosted services to a limited number of people behind a firewall.

Cloud model – Hybrid cloud



- An organization provides and manages some resources in-house and has others provided externally.
- For example: public cloud service for archived data, and in-house storage (private cloud) for operational customer data
- Ideally, it allows to take advantage of the scalability and cost-effectiveness that a public cloud offers without exposing mission-critical applications and data to third-party vulnerabilities.

Cloud model – Public vs. Private

Public Cloud service

Advantages:

- Simplicity and efficiency
- Low cost
- Reduced time
- No maintenance
- No contracts with HW suppliers

Disadvantages:

- Lack of direct control
- Slow/Limited speed
- Lack of investment
- Perceived weaker security

Private Cloud service

Advantages:

- Greater/Complete control
- Potentially increased security
- Higher performance
- Deeper compliance
- Customizable/Manageable

Disadvantages:

- Higher cost
- On-site maintenance
- Capacity ceiling

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VPS

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Virtual Private Servers (VPS)

- High performance at unbeatable prices
- Simplicity, scalability and customisation
- Unlimited traffic and up to 2 Gbit/s bandwidth*



VALUE

\$6.00/month

\$5.52

/month

12 months commitment (-8%)

[Order now](#)

ESSENTIAL

\$11.50/month

\$10.58

/month

12 months commitment (-8%)

[Order now](#)

COMFORT

\$23.00/month

\$21.16

/month

12 months commitment (-8%)

[Order now](#)

ELITE

From \$24.00/month

\$31.28

/month

[Order now](#)

8 vCore

From 8 GB to 32 GB

From 160 GB to 510 GB SSD NVMe

OVH Cloud: World presence



OVH Cloud: Datacenters



RBX-1



RBX-2



RBX-3



RBX-4



RBX-5



SBG-1



SBG-2



GRA-1

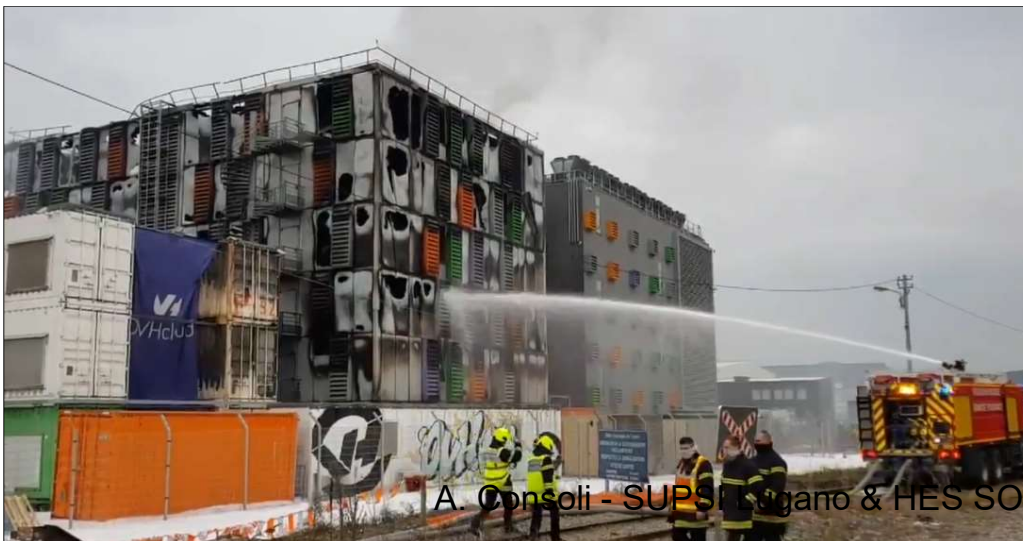


BHS-1



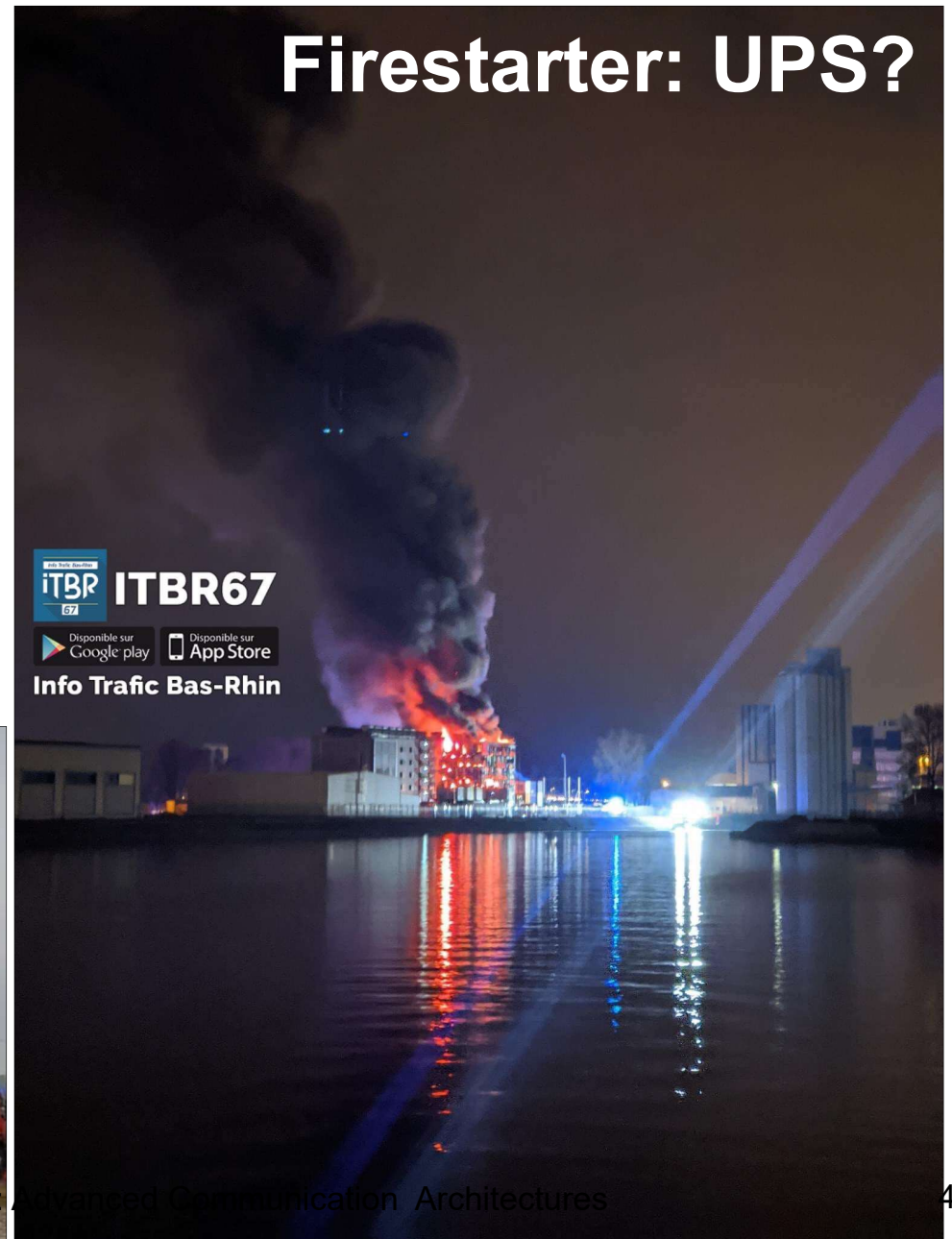
BHS-2

OVH Cloud Fire Incident



Firestarter: UPS?

ITBR
67
Disponible sur
Google play
Disponible sur
App Store
Info Trafic Bas-Rhin



OVH Cloud Fire Incident

 **Octave Klaba** ✓ @olesovhcom · 10 mars 2021

We have a major incident on SBG2. The fire declared in the building. Firefighters were immediately on the scene but could not control the fire in SBG2. The whole site has been isolated which impacts all services in SGB1-4. We recommend to activate your Disaster Recovery Plan.

 **Octave Klaba** ✓
@olesovhcom

Update 5:20pm. Everybody is safe.
Fire destroyed SBG2. A part of SBG1 is destroyed.
Firefighters are protecting SBG3. no impact SBG4.

5:24 AM · 10 mars 2021

♥ 859 💬 419 🔗 Copier le lien du Tweet



Chers clients,

nous avons reçu cette nuit, à 23h42, une remontée d'informations de l'**indisponibilité de notre serveur dédié**, hébergé dans le datacenter d'OVH à Strasbourg.

Ce matin, à 3h53, nous avons été avertis que cette indisponibilité provient d'un **incendie qui s'est déclaré dans la salle de serveurs** où se trouve le nôtre.

Les pompiers sont intervenus immédiatement mais n'ont pas réussi à maîtriser l'incendie du bâtiment SBG2. Par mesure de sécurité, **toute l'alimentation électrique du datacenter a été stoppée**.

Actuellement, tous les sites internet dont nous avons la gestion sont indisponibles.

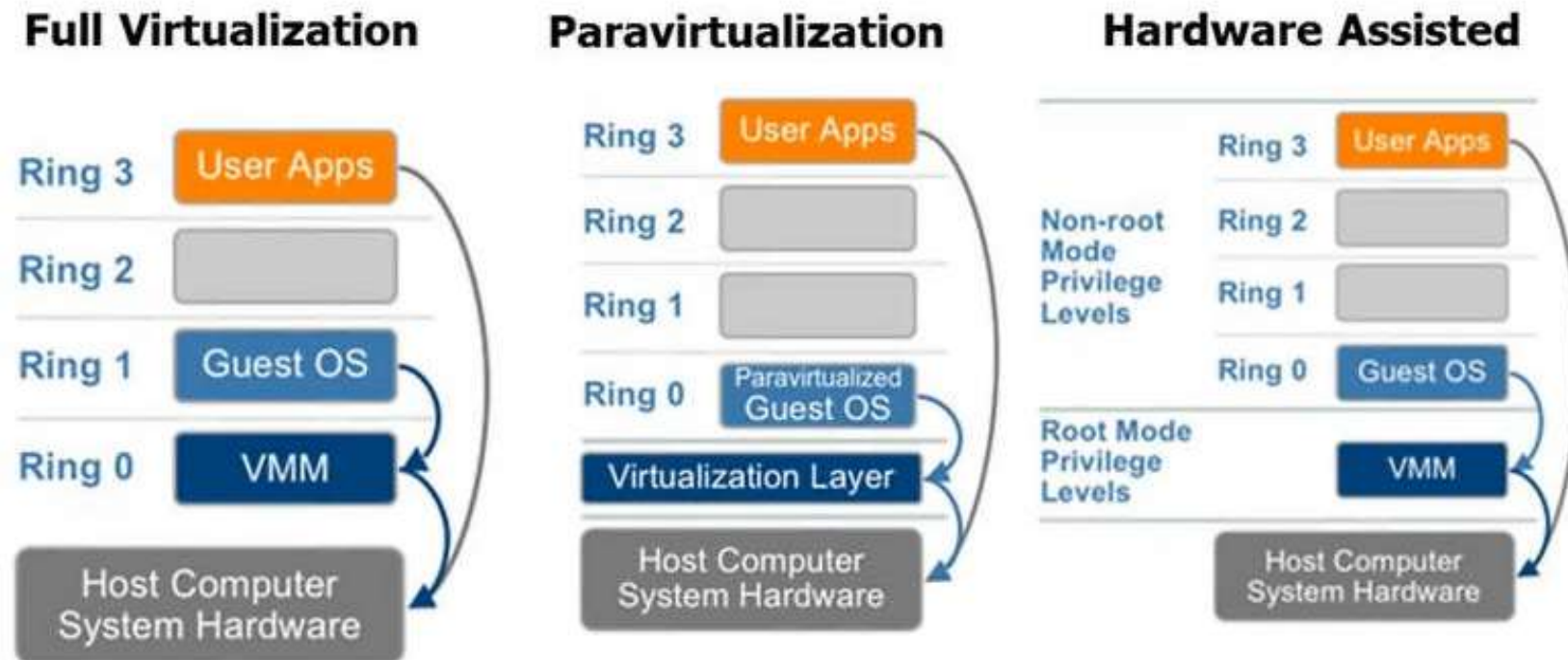
Nous sommes en train d'analyser l'infrastructure afin de voir si d'autres services sont également impactés (serveurs mail, téléphonie IP...).

Nous vous tiendrons informés sur l'évolution de cet incident et les mesures qui seront prises au niveau du datacenter afin de rétablir au plus vite tous nos services.

Virtualization

Virtualization vs. Paravirtualization vs. Hardware Assisted

Assisted Virtualization: Architectural comparison



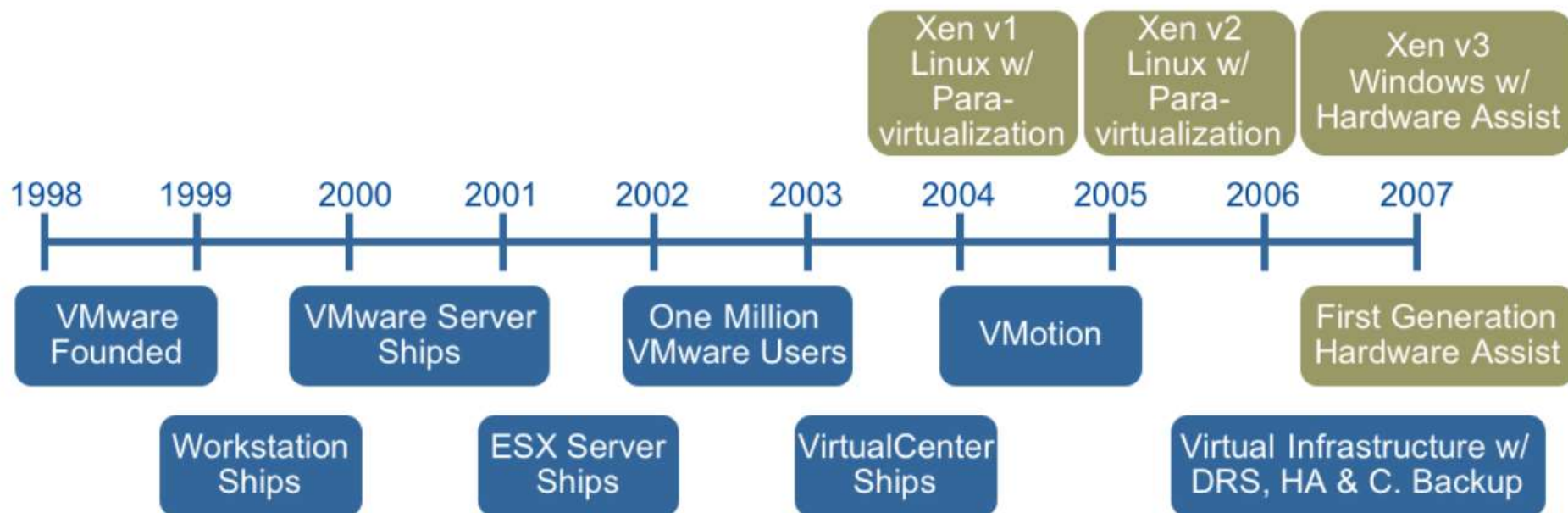
<https://zhihuicao.wordpress.com/2015/06/13/para-virtualization-full-virtualization-differences>

Understanding Full Virtualization, Paravirtualization, and Hardware Assist

Virtualization vs. Paravirtualization vs. Hardware Assisted Virtualization

- **Paravirtualization** is virtualization in which the guest operating system (the one being virtualized) is aware that it is a guest and accordingly has drivers that, instead of issuing hardware commands, simply **issues commands directly to the host operating system**. This will include things such as memory management as well.
- **Full Virtualization** is virtualization in which the guest operating system is unaware that it is in a virtualized environment, and therefore **hardware is virtualized by the host operating system** so that the guest can issue commands to what it thinks is actual hardware, but really are just simulated hardware devices created by the host.
- **Hardware Assisted Virtualization** is a type of Full Virtualization where the microprocessor architecture has **special instructions to aid the virtualization of hardware**. These instructions might allow a virtual context to be setup so that the guest can execute privileged instructions directly on the processor, even though it is virtualized. If said instructions do not exist, Full Virtualization is still possible, however it must be done via software techniques such as **Dynamic Recompilation** where the host recompiles on the fly privileged instructions in the guest to be able to run in a non-privileged way on the host.
- There is also a combination of Para Virtualization and Full Virtualization called **Hybrid Virtualization** where parts of the guest operating system use paravirtualization for certain hardware drivers, and the host uses full virtualization for other features. This often produces superior performance on the guest without the need for the guest to be completely paravirtualized. An example of this: The guest uses full virtualization for privileged instructions in the kernel but paravirtualization for IO requests using a special driver in the guest. This way the guest operating system does not need to be fully paravirtualized (which is sometimes not available) but can still enjoy some paravirtualized features.

Timeline of x86 virtualization technologies



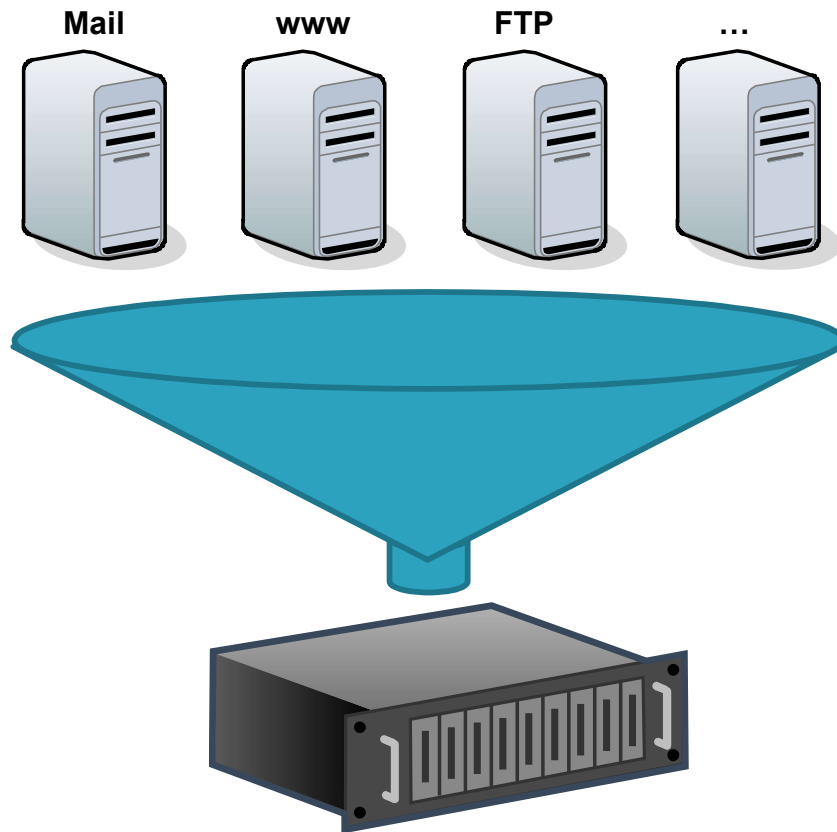
Understanding Full Virtualization, Paravirtualization, and Hardware Assist

Virtualization vs. Paravirtualization vs. Hardware Assisted Virtualization: Products

Logo	Name	Type
	KVM	Full virtualization
	QEMU	Powerful emulator
	Xen	Paravirtualized & Hardware Assisted
	VMware	Full virtualization, Paravirtualized & Hardware Assisted
	ProxMox	Full virtualization, Paravirtualized & Hardware Assisted

Virtualization vs. Cloud – 1

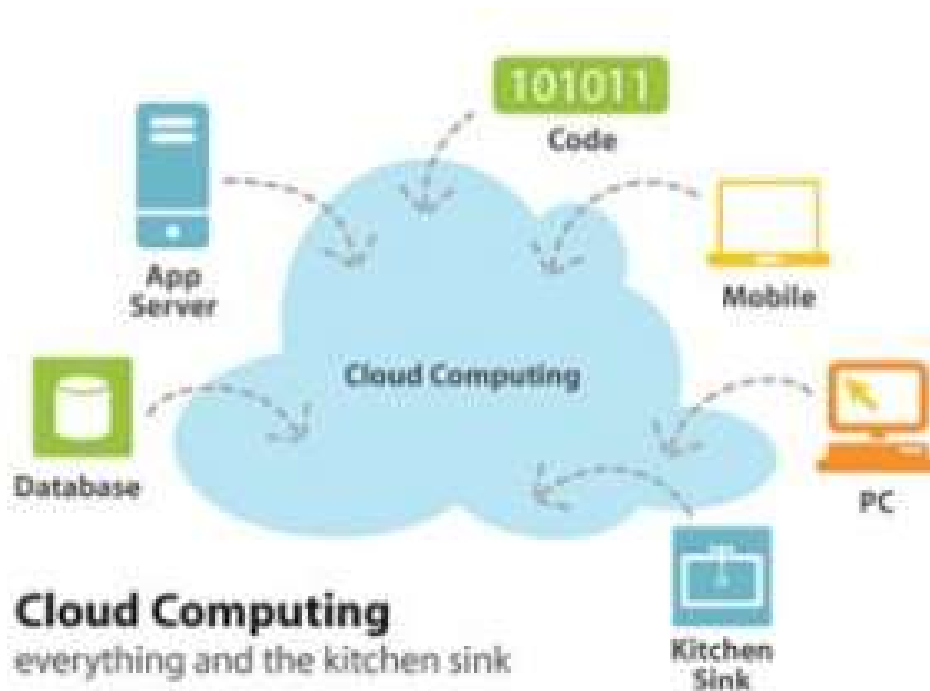
Virtualization: More servers on the same hardware



- Virtualization allows to run several servers on the same hardware.
- Most virtualization systems allow the virtual servers to be easily moved from one physical host to another.
- Greatest benefit: reducing the cost of that infrastructure for companies by maximizing the usage of the physical resources.

Virtualization vs. Cloud – 2

Cloud computing: Measured resources, pay for what you use

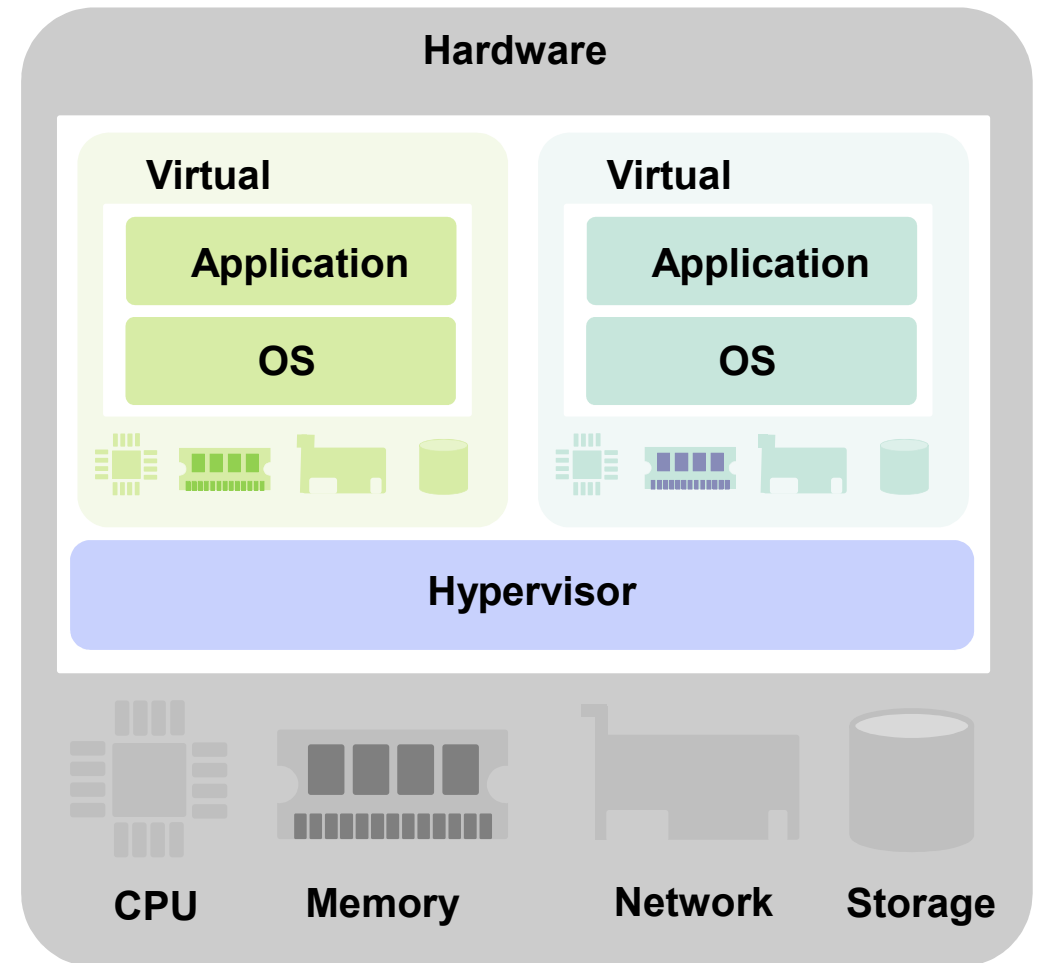


- Cloud computing grew out of the concept of “Utility computing”
- Distribution across many servers provides:
 - Redundancy (even geographic redundancy)
 - High availability
 - Flexibility
 - Scalability
- Drawback: You do not control the servers

Virtualization – Virtual machines

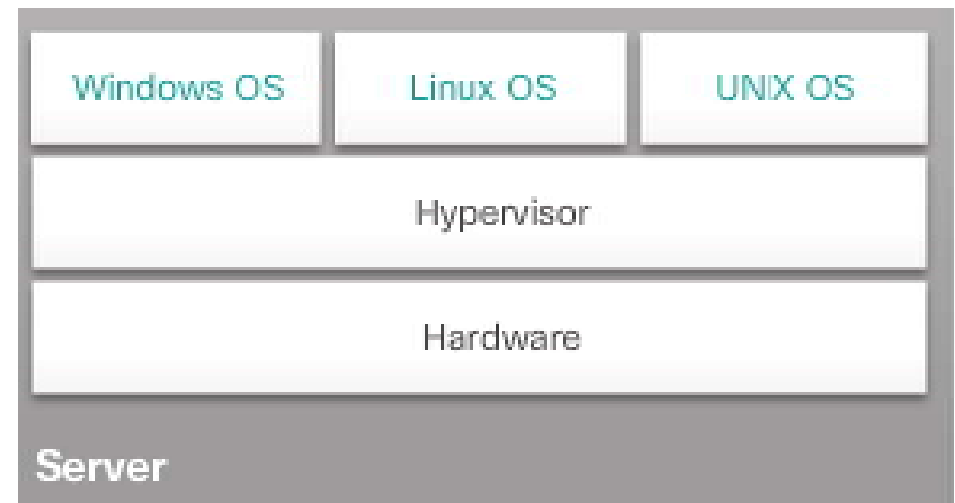
Virtual machines:

- creation of multiple virtual machines on a physical machine (computer).
- physical hardware is shared between virtual machines.
- Hypervisor translates access to virtual hardware to access to the physical hardware.



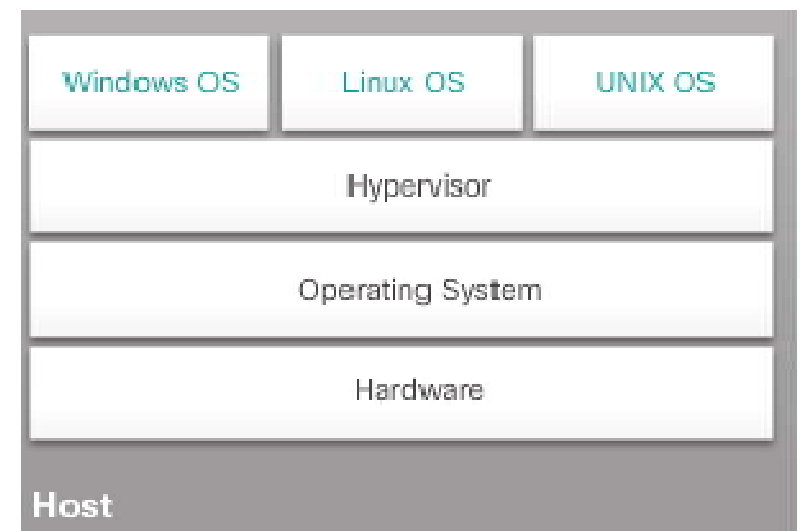
Type 1 Hypervisors

- Type 1 hypervisors are also called the “bare metal” approach because the hypervisor is installed directly on the server or networking hardware.
- They have direct access to the hardware resources; therefore, they are more efficient than hosted architectures.
- They require a “management console” to manage the hypervisor.
- Popular Type 1 hypervisors include:
 - Citrix XenServer
 - Microsoft Hyper-V 2008/2012
 - Oracle VM Server for x86/SPARC
 - VMware ESXi



Type 2 Hypervisors

- A hypervisor is a software that creates and runs VM instances.
- Type 2 hypervisor is installed on the existing OS, such as Mac OS X, Windows, or Linux.
- The computer, on which a hypervisor is supporting one or more VMs, is a host machine.
- Type 2 hypervisors are also called hosted hypervisors.
- Common Type 2 hypervisors include:
 - Virtual PC
 - VMware Workstation
 - Oracle VM VirtualBox
 - VMware Fusion
 - Mac OS X Parallels



Virtualization vs. Cloud – 3

Which one is right for my application?

➤ **No unique answer!!**

- One key consideration is: **When do you need to save the money?**
 - With virtualization: Great deal of upfront cost
 - With Cloud computing: A new application may not need many resources initially. However, as your application becomes popular and uses more resources, paying by the resource may become more expensive.
- Another important consideration is: **How safe will your data be?**

Cloud – Glossary

Glossary

- <https://whatis.techtarget.com/glossary/Cloud-computing>
- <https://https://www.webopedia.com/reference/cloud-computing-guide/>

References

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